

Package Peak Temperature (°C)

- 3D (all-DRAM): DRAM tier T
- xBM: DRAM tier T
- small box = peak GPU T

thermal headroom

3D thermal penalty

GPU frequency to stay thermally viable

Config	GPU freq	Through-put	Thr. density
3D all-DRAM	0.5×	72% (87%)	122% (146%)
xBM	0.8×	89% (115%)	150% (193%)

first = ref HBM bandwidth · () = expected with 4× bandwidth (3D)

Note: identical thermal boundaries for all cases.

STCO approach

2.5D
69.1°C

(paper ref.)

3D
121.2°C
GPU 143.1°C

HBM base die removal

3D
121.3°C
GPU 139.8°C

HBM stack merging

xBM
100.3°C
GPU 137.7°C

xBM
100.4°C
GPU 134.4°C

3D
111.4°C
GPU 123.0°C

Top die thinning

3D
110.1°C
GPU 121.2°C

GPU frequency scaling

3D: 0.5×

xBM
93.1°C
GPU 117.4°C

xBM
91.9°C
GPU 115.7°C

xBM: 0.8×

Thermal silicon optimization

3D
91.2°C
GPU 99.0°C

xBM
85.6°C
GPU 106.7°C

3D
83.8°C
GPU 86.6°C

xBM
79.7°C
GPU 91.7°C